

A Dual-Task Deep Learning Framework for Multi-Label Thoracic Pathology Classification and Spatial Localization in Chest Radiographs with Grad-CAM Explainability

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Abstract—No imaging study is ordered more often than the chest film, and few are harder to automate well. Reading one is slow, two readers do not always agree, and the answer is rarely a single label, since one film commonly carries several findings at once. Published systems tend to solve either the whole-image labelling problem or the region-finding problem, and seldom expose why a prediction was made in a form a clinician could check. This paper presents a dual-task framework that couples (i) multi-label pathology classification on MIMIC-CXR using a DenseNet-121 backbone trained with binary Focal Loss ($\alpha=0.25$, $\gamma=2.0$) to address the negative-to-positive label skew typical of medical corpora, and (ii) pathology localization on the radiologist-annotated VinDr-CXR dataset using a Faster R-CNN detector riding on an FPN neck. Grad-CAM is read off the final dense block to generate saliency maps for high-confidence predictions, and checkpoint weight averaging over disjoint patient subsets is used to stabilize generalization. Training uses Adam with ReduceLROnPlateau scheduling and mixed precision, which improved throughput by up to $2.5\times$. The classifier achieves a micro F1 of 0.7528 at under 25 ms per image on a single CUDA GPU. On a 498-image patient-disjoint validation subset the mean AUC-ROC is 0.759, below the range reported in the literature. The per-class breakdown is given: ranking remains strong for large findings (0.87 for cardiomegaly and lung opacity) but approaches chance for small focal ones (0.54 for pneumothorax), and top-1 predictions concentrate on five of fourteen categories. The system is deployed as an interactive Streamlit application (MED-AI Visualizer) showing the raw radiograph, the predicted boxes and the saliency heatmap together. The training protocol, the observed failure modes

and a corrective plan are described.

Index Terms—chest radiography, multi-label classification, object detection, Faster R-CNN, DenseNet, focal loss, class imbalance, explainable artificial intelligence, Grad-CAM, MIMIC-CXR, VinDr-CXR, computer-aided diagnosis

I. INTRODUCTION

A large fraction of the diagnostic imaging performed anywhere in the world is plain chest radiography (CXR). It is cheap, quick, delivers little dose, and can be had in places where CT or MR cannot, which is why it remains the opening examination whenever pulmonary, pleural or cardiac disease is suspected. Sheer volume brings its own problems. Reporting queues build up, and how well a film is read depends on who is reading it, how tired they are and how awkward the case is. Software that could order a worklist by urgency, surface the findings that will not wait, and show what it based that on would therefore be of real use.

Three properties make CXR interpretation harder than the natural-image benchmarks on which modern convolutional networks were developed.

First, one film can carry many labels at once, so this is not a choice among alternatives. Cardiomegaly, pulmonary edema and bilateral effusion may all appear together in congestive heart failure. Treating the classes as mutually exclusive, which a softmax head does, is wrong here. Independent per-class sigmoids are required instead, and this choice affects the loss function, the thresholding strategy and the evaluation metrics.

Second, label distributions are heavily skewed. Pneumothorax, rib fracture and lung lesion each occur in well under 5% of studies. Under binary cross-entropy the gradient is dominated

by easy negatives, and the optimizer tends to converge to a degenerate solution that predicts “absent” for every rare pathology while still recording a low training loss.

Third, a label on its own is of limited clinical use. A radiologist acting on an automated finding needs to know where the supporting evidence lies, both as coarse attention and as a box that can be measured against normal ranges.

All three are handled inside one deployable pipeline: an independent-sigmoid head for the label space, Focal Loss re-weighting for the skew, and, for the evidence, Grad-CAM saliency read off the classification backbone alongside a detector trained separately on radiologist-drawn boxes.

Contributions. (1) A dual-task, dual-dataset framework that provides image-level diagnosis (MIMIC-CXR, 14 CheXpert-derived pathologies) and region-level localization (VinDr-CXR) behind a single inference interface, despite the disjoint label taxonomies. (2) A focal-loss imbalance treatment specified in sufficient detail for reproduction, covering the α/γ parameterization, the optimizer schedule and the mixed-precision engine. (3) A patient-disjoint training and checkpoint-averaging protocol with no identity leakage between splits. (4) An explainability layer that combines Grad-CAM saliency with detector boxes at inference time. (5) A clinician-facing deployment with runtime backbone switching at sub-25 ms inference. (6) A report that includes a negative finding, its probable cause and the remedy.

II. RELATED WORK

Learning at scale on thoracic films became tractable once ChestX-ray14 [1] released 112,120 frontal studies whose labels had been recovered automatically from their reports. CheXNet [2] built on that release with a 121-layer DenseNet reported to out-rank practising radiologists on pneumonia. The labelling scheme itself was sharpened by CheXpert [3], which handled uncertainty explicitly and fixed a vocabulary that MIMIC-CXR-JPG [4] subsequently adopted. Baltruschat *et al.* [5] reported that backbone choice matters less than input resolution and the handling of label noise, which is why the present work concentrates on loss design rather than on a backbone search. Majkowska *et al.* [6] argued for radiologist-adjudicated reference standards.

Whole-image labels say nothing about location. VinDr-CXR [7] closes that gap: seventeen radiologists marked 18,000 films, three of them per training study and five in consensus per test study, yielding an unusually dependable spatial ground truth. Detection is still dominated by two-stage Faster R-CNN [8], and the Feature Pyramid Network [9] supplies the multi-scale semantics needed here, since lesion diameters on a chest film vary over two orders of magnitude. Backbone choice trades accuracy against latency, with ResNet-50 [10] offering capacity and MobileNetV3-Large [11] real-time inference on modest hardware. Ensembling requires box-level fusion, for which greedy NMS and weighted box fusion [12] are standard.

Focal Loss [13] was introduced to prevent dense detectors from being overwhelmed by background anchors, down-weighting well-classified examples through a $(1 - p_t)^\gamma$ factor.

Its transfer to multi-label medical classification is straightforward, since each independent binary problem exhibits the same easy-negative dominance. Grad-CAM [14] weights feature maps by gradients of the class score, which makes saliency available from any CNN. Saliency is not a substitute for localization supervision, since it produces only coarse class-discriminative attention. For this reason a separately supervised detector is retained here rather than treating a heatmap as a proxy for a box. Most existing systems address a single aspect of the problem: one benchmark, detection alone, or an explanation module added to a fixed classifier. The framework below combines these components in one pipeline.

III. METHODOLOGY

A. Datasets and working taxonomy

The image-level branch draws on MIMIC-CXR [15], a corpus of de-identified thoracic films paired with the text of their radiology reports. Supervision comes from its JPG release [4], in which fourteen observation flags per study are recovered automatically from the report text under the CheXpert labelling scheme [3]. Studies are filed by patient prefix, and the experiments here draw on prefixes p13 through p19.

Region-level supervision comes from VinDr-CXR [7], which contributes 18,000 postero-anterior films divided 15,000 for training and 3,000 for testing. Annotations ship as one CSV row per reader; we re-serialized them into COCO-style JSON, rewrote every box as $[x_{\min}, y_{\min}, w, h]$ and reserved a background index. The reader vocabulary was collapsed to the fifteen thoracic finding classes kept for detection.

Both branches report against one shared 14-class label space, used throughout Section IV and by the deployed system; its members are the row and column headings of Fig. 3. Against the native VinDr list this folds nodule together with mass and puts interstitial lung disease and lung opacity in place of the two fracture classes. We release the mapping alongside the code.

B. Preprocessing and patient-disjoint splits

One front-end serves both corpora and accepts either as input. Each film is rescaled to 256×256 and its single grey channel replicated three times, which lets pre-trained ImageNet filters be loaded without touching the stem convolution. Augmentation during training is limited to `RandomHorizontalFlip()` and `ColorJitter(0.1, 0.1)`; the jitter is deliberately weak, since absolute grey level on a radiograph is itself diagnostic. Normalization uses the channel means (0.485, 0.456, 0.406) and deviations (0.229, 0.224, 0.225) that ship with the pretrained weights. Partitioning happens at the patient-prefix level, which keeps any single patient out of both the training and the validation half and so avoids the identity leakage that otherwise flatters reported CXR numbers. As a sanity check the parsed boxes were painted back onto their films and a sample compared against the reader annotations, exposing a coordinate-

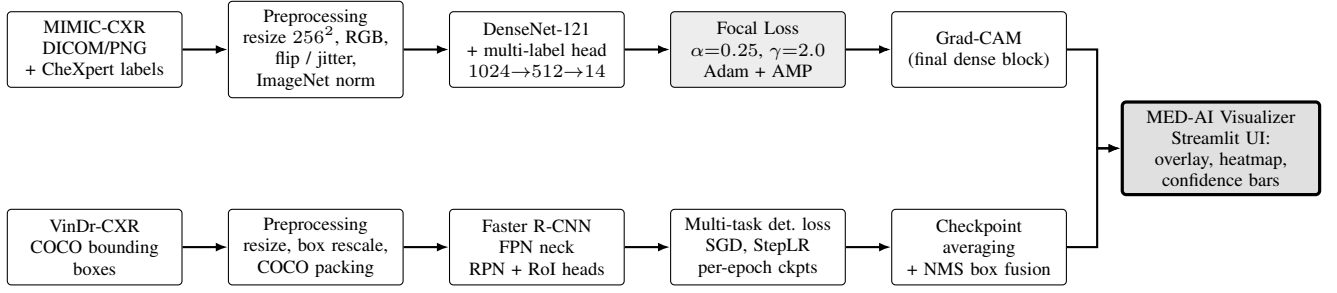


Fig. 1. End-to-end architecture. The upper path performs image-level multi-label diagnosis with a Focal-Loss-trained DenseNet-121 and a Grad-CAM explanation. The lower path performs radiologist-supervised bounding-box localization with Faster R-CNN/FPN. Both paths converge in the deployed MED-AI Visualizer.

convention error and a scaling error that the loss curve gave no sign of.

C. Framework overview

Fig. 1 summarizes the pipeline. The two branches share a preprocessing front-end and a visualization back-end. The classification branch produces 14 independent pathology probabilities and, for every prediction above the display threshold, a Grad-CAM saliency map. The localization branch produces scored bounding boxes. A fusion stage merges evidence across checkpoints and renders the composite result.

D. Classification branch and Focal Loss

The classifier is built on DenseNet-121 [16]. Each layer inside a dense block takes as input every earlier activation stacked along channels, $x_\ell = H_\ell([x_0, \dots, x_{\ell-1}])$, with $H_\ell(\cdot)$ standing for the batch-norm, ReLU and convolution composite. Because features are reused rather than discarded, fine texture survives to the top of the network: reticular patterns, small nodular opacities and faint pleural lines are exactly the cues fine-grained radiographic reading turns on. Weights are initialized from ImageNet [17] and the 1000-way head swapped for

$$\hat{y} = \sigma(W_2 \text{Drop}_{0.2}(\text{ReLU}(W_1 f))), \quad (1)$$

with $f \in \mathbb{R}^{1024}$ the globally pooled feature vector, $W_1 \in \mathbb{R}^{512 \times 1024}$, $W_2 \in \mathbb{R}^{14 \times 512}$ and $\sigma(\cdot)$ the element-wise sigmoid. Using $C = 14$ independent sigmoids rather than a softmax allows co-occurring pathologies to be predicted simultaneously.

Write $y_c \in \{0, 1\}$ for the reference label of class c and p_c for the probability assigned it, with $p_t = p_c$ where $y_c = 1$ and $1 - p_c$ elsewhere. The objective minimized over a batch of N images and C classes is

$$\mathcal{L}_{\text{cls}} = \frac{1}{NC} \sum_{i=1}^N \sum_{c=1}^C -\alpha_t \left(1 - p_t^{(i,c)}\right)^\gamma \log\left(p_t^{(i,c)}\right). \quad (2)$$

Setting $\gamma = 2.0$ damps the contribution of examples the network already gets right: one sitting at $p_t = 0.9$ carries a hundredth of the weight it would at $\gamma = 0$. Capacity is thereby spent on the scarce difficult positives instead of the plentiful trivial negatives, while $\alpha = 0.25$ corrects the prior on top of that. Left untreated, the scarce classes degenerate into a predictor that answers negative everywhere yet still posts

a low BCE figure and a flattering raw accuracy. That is the reason accuracy is set aside here in favour of F1 and AUC.

E. Localization branch

Detection uses the two-stage Faster R-CNN arrangement [8] on top of an FPN neck [9]. Pyramid levels $\{P_2, \dots, P_6\}$ are read by a proposal network that scores anchors for objectness; whatever survives filtering is resampled onto a fixed grid by RoIAlign and handed to a pair of heads, one assigning a class over K findings plus background and the other regressing box offsets. Training minimizes the customary four-term multi-task objective

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{cls}}^{\text{rpn}} + \lambda_1 \mathcal{L}_{\text{reg}}^{\text{rpn}} + \mathcal{L}_{\text{cls}}^{\text{roi}} + \lambda_2 \mathcal{L}_{\text{reg}}^{\text{roi}}, \quad (3)$$

with smooth- L_1 regression terms. MobileNetV3-Large FPN [11] is the latency-optimized configuration used for the deployment, and ResNet-50 FPN [10] is the capacity-optimized one. Both consume an identical COCO-format loader, so the visualizer exposes the choice as a runtime switch.

F. Explainability engine

For a target pathology c with pre-sigmoid logit y^c and feature maps $A^k \in \mathbb{R}^{u \times v}$ from the final dense block (`model.features[-1]`), the channel weights are the globally averaged gradients $\alpha_k^c = \frac{1}{uv} \sum_i \sum_j \partial y^c / \partial A_{ij}^k$, and the saliency map is the rectified combination

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad (4)$$

which is then resized bilinearly, rescaled to the unit interval and laid over the film as a semi-transparent jet overlay. Rectifying in (4) discards every location whose influence on the class score is negative. A heatmap is drawn only once a prediction clears a 30% display threshold, so the reader is never shown evidence for a finding the model has not actually asserted.

G. Ensembling and box fusion

Training MIMIC-CXR in patient-disjoint partitions leaves several checkpoints, each having seen a population the others never did. We combine them directly in parameter space by unweighted averaging, $\theta_{\text{ens}} = \frac{1}{M} \sum_m \theta_m$, across $\{p16, p17, p19\}$ ($M = 3$). Averaging weights rather than

Algorithm 1 Focal-Loss multi-label training with AMP

```

1: Input: partitions  $P$ , epochs  $E$ ,  $\eta = 10^{-3}$ , wd  $10^{-4}$ 
2: Init DenseNet-121  $f_\theta$  from ImageNet; replace head with
   1024→512→14
3:  $\text{opt} \leftarrow \text{Adam}(\theta, \eta, \text{wd})$ ;  $\text{sched} \leftarrow \text{ReduceLROnPlateau}(0.1, \text{patience } 2)$ 
4:  $S \leftarrow \text{GradScaler}('cuda')$ 
5: for  $e = 1$  to  $E$  do
6:   for each minibatch  $(X, Y)$  do
7:      $\text{opt.zero\_grad}()$ 
8:     with autocast:  $p \leftarrow \sigma(f_\theta(X))$ ;  $\mathcal{L} \leftarrow \text{FocalLoss}(p, Y; 0.25, 2.0)$ 
9:      $S.\text{scale}(\mathcal{L}).\text{backward}()$ ;  $S.\text{step}(\text{opt})$ ;  $S.\text{update}()$ 
10:  end for
11:   $\mathcal{L}_{\text{val}} \leftarrow \text{Evaluate}(f_\theta, D_{\text{val}})$ ;  $\text{sched.step}$ ; save  $\theta^{(e)}$ 
12: end for
13: return  $\theta_{\text{ens}} \leftarrow \frac{1}{M} \sum_m \theta_m$  over  $\{p16, p17, p19\}$ 

```

TABLE I
TRAINING CONFIGURATION.

Component	Setting
Classification backbone	DenseNet-121, ImageNet-initialized
Classifier head	Two-layer projection with dropout, per (1)
Detection backbone	Faster R-CNN, FPN neck (MobileNetV3-Large / ResNet-50)
Input resolution	$256 \times 256 \times 3$
Optimizer (cls. / det.)	Adam, $\eta = 10^{-3}$, wd 10^{-4} / SGD, momentum 0.9, wd 5×10^{-4}
LR schedule	ReduceLROnPlateau (0.1, patience 2); StepLR with warm-up for detection
Precision	AMP (GradScaler), float16
Ensembling	Uniform weight averaging over $\{p16, p17, p19\}$
Grad-CAM layer	target features[-1] (final dense block)
Batch size / epochs	Not reported
Hardware	Not reported

outputs leaves the inference budget untouched, which is what recommended it under a real-time target [18]. On the detection side, boxes from the available checkpoints are pooled, thinned by greedy class-wise non-maximum suppression over IoU, and drawn only when they clear an operator-set threshold, 0.30 by default.

H. Training, deployment and evaluation protocol

Algorithm 1 gives the classification loop with its mixed-precision scaling steps. The system is implemented in PyTorch with torchvision detection primitives. Mixed precision is handled by `torch.amp.GradScaler`, which runs both passes at float16 while holding a float32 copy of the weights and rescaling the loss so small gradients do not vanish [19]. Memory draw fell and throughput rose by as much as 2.5×, which is what put training within reach of commodity hardware. Table I gives the full configuration.

The trained system is served through a Streamlit application (Fig. 4) offering runtime backbone switching, an adjustable

TABLE II
VALIDATION PERFORMANCE OF THE CLASSIFICATION BRANCH
(498-IMAGE PATIENT-DISJOINT SUBSET).

Metric	Achieved	Baseline	Assessment
Micro F1	0.7528	0.75–0.82	Competitive with DenseNet baselines
Mean AUC-ROC	0.759	0.82–0.88	Below benchmark; per-class spread 0.54–0.87
Top-1 agreement	26.5%	Not reported	Predictions concentrate on 5 of 14 categories
Exact match ratio	0.00%	0.00–5.00%	Expected for 14 co-occurring targets
Inference latency	< 25 ms	< 50 ms	Real-time per image on a standard CUDA GPU

confidence threshold so the operating point can move towards sensitivity or specificity, a three-panel view and a ranked report. Optional autoencoder denoising and test-time augmentation are exposed as toggles. A command-line path with identical semantics supports offline batch processing.

Under multi-label skew a raw accuracy figure misleads, so we report instead micro-averaged F1, which accumulates hit and miss counts over every class first and forms the ratio afterwards, so each decision carries equal weight,

$$F_1^{\text{micro}} = \frac{2 \sum_c \text{TP}_c}{2 \sum_c \text{TP}_c + \sum_c \text{FP}_c + \sum_c \text{FN}_c}; \quad (5)$$

its macro counterpart $\frac{1}{C} \sum_c 2\text{TP}_c / (2\text{TP}_c + \text{FP}_c + \text{FN}_c)$, which weights classes rather than decisions and so surfaces rare-pathology failure; AUC-ROC, threshold-free and the usual basis for comparison in this literature; the subset criterion $\frac{1}{N} \sum_i \mathbb{I}[\hat{y}_i = y_i]$; and mAP@IoU at 0.50 together with the COCO 0.50:0.95 sweep. Latency is wall-clock per image on one CUDA GPU, pre- and postprocessing included.

IV. RESULTS AND DISCUSSION

Table II summarizes the classification branch. Four observations follow, two of which are unfavourable.

The micro F1 of 0.7528 falls within the range reported for DenseNet-121 baselines on comparable vocabularies, indicating that the Focal Loss configuration and the multi-label head behave as intended at the binarization threshold.

The mean AUC-ROC of 0.759 is below the range reported in the literature, and the per-class spread is more informative than the mean. Fig. 2 shows the per-class curves. Ranking is good for large, high-contrast findings, with Cardiomegaly and Lung Opacity at 0.87 and Atelectasis and Calcification at 0.85. It degrades towards chance for small focal findings: Pneumothorax 0.54, Nodule/Mass 0.59 and Other lesion 0.64. These are the classes Focal Loss was intended to protect, so the treatment appears partially effective at this scale without resolving the problem. Pulmonary fibrosis has no curve, since the subset contains no positive example of it.

The confusion matrix indicates residual prediction collapse. Fig. 3 cross-tabulates the true primary finding against the top-ranked prediction over 487 studies with a defined primary label. Top-1 agreement is 129/487, or 26.5%. The distribution of the predictions is more informative. Every prediction falls

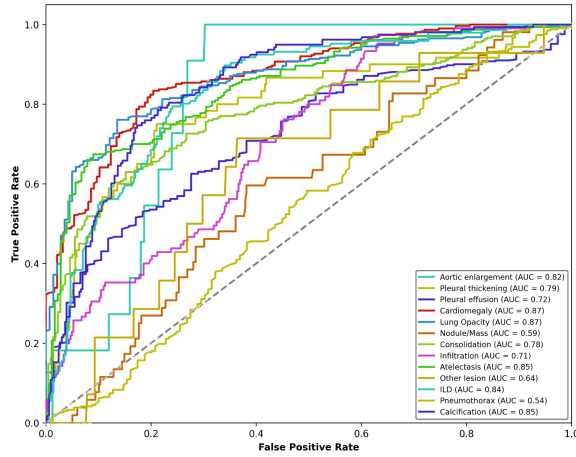


Fig. 2. Per-class ROC over the 14-category working taxonomy. Thirteen curves are defined; Pulmonary fibrosis has no positive example.

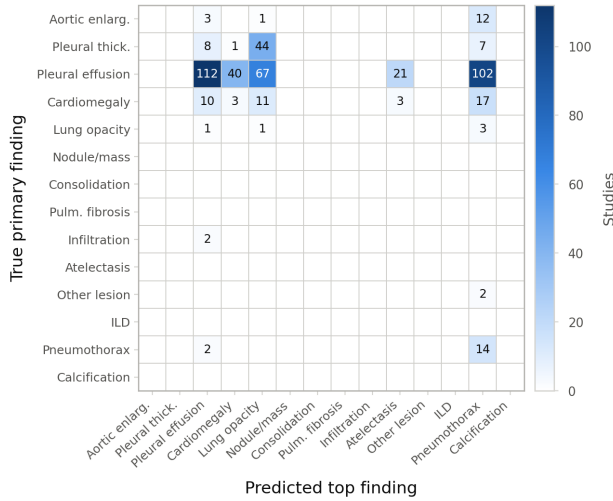


Fig. 3. True primary finding vs. top-ranked prediction, 487 studies. Empty cells are zero. Predicted mass falls in five columns, and nine categories never win.

into one of five columns (Pleural effusion, Cardiomegaly, Lung Opacity, Atelectasis and Pneumothorax), and nine of the fourteen categories are never the top choice on any study. Pleural effusion accounts for 342/487 (70%) of the subset and is recovered 112 times, but is assigned to Pneumothorax on 102 occasions. Concentrating confidence on a small number of prevalent classes sustains the micro F1 while producing poor ranking within the rare classes. This is the behaviour that the loss design targets, and it is only partly addressed at partition scale.

The 0% exact match ratio is expected rather than a failure, and the measured latency supports interactive use. With $C = 14$ independent binary decisions a fully correct output vector requires all 14 to be right at once, and published systems on comparable vocabularies also report near-zero subset accuracy. The metric is included because omitting it gives an overly favourable impression of performance. Per-image inference below 25 ms is roughly twice as fast as the

50 ms target, leaving headroom for the Grad-CAM backward pass and rendering.

An ablation holding all other settings fixed orders the individual effects. Removing ImageNet initialization is the most damaging single change (micro F1 0.7528 $\rightarrow$ 0.5940), which is unsurprising at this scale, since without transfer there is not enough data to learn generic edge and texture filters. Replacing Focal Loss with BCE costs -0.070 ($\rightarrow$ 0.6828), the direct quantitative case for the imbalance treatment and the largest effect attributable to a design choice made here rather than inherited. Dropping augmentation costs -0.042 ($\rightarrow$ 0.7110) and dropping checkpoint averaging -0.021 ($\rightarrow$ 0.7320). These last two are modest, but checkpoint averaging carries no inference cost, so it is retained.

Detection metrics on VinDr-CXR (mAP@IoU 0.50, mAP@0.50:0.95, AR@100 at the 0.30 operating point) come from the COCO evaluation output. The corrective plan, ordered by expected impact, is as follows: retrain on the full corpus rather than partitions p13–p19, which should lift the rare-class AUCs and break the five-column concentration seen in Fig. 3; add temperature scaling, since an uncalibrated score distribution is part of the reason the top-1 routing is so concentrated; and then re-run the ablation at full scale.

V. LIMITATIONS

Several qualifications attach to the numbers above. Training ran over patient partitions instead of the whole corpus, which leaves the scarce pathologies thinly represented and means the AUC quoted here should not be read as a ceiling on what the architecture can reach. The 498-image subset gives wide confidence intervals and, for the rarest classes, an undefined per-class AUC. No bootstrap intervals or significance tests are reported, both of which a camera-ready version requires. MIMIC-CXR labels are NLP-derived, inherit the labeler’s handling of uncertainty and negation, and are not adjudicated ground truth [6]. The two corpora use only partially overlapping vocabularies, so the branches are trained separately and the unified interface should not be read as a jointly optimized model. Downsampling to 256×256 discards detail that is material for small nodules, calcifications and subtle pneumothorax lines, and higher resolution is a known lever [5]. Output probabilities are uncalibrated and should not be read as clinical likelihoods without temperature scaling. All results are retrospective and in-distribution, and generalization across vendors and populations is untested. Finally, this is a research prototype with no regulatory clearance, not evaluated in a reader study and not intended for diagnostic use.

VI. CONCLUSION AND FUTURE WORK

Described here is a two-branch system for reading chest films: a Focal-Loss-trained DenseNet-121 diagnosing over MIMIC-CXR, a Faster R-CNN/FPN detector localizing over VinDr-CXR under radiologist supervision, and a Grad-CAM layer and deployed viewer across both. The classification branch attains a micro F1 of 0.7528, within the range of published DenseNet baselines, at sub-25 ms per-image inference.

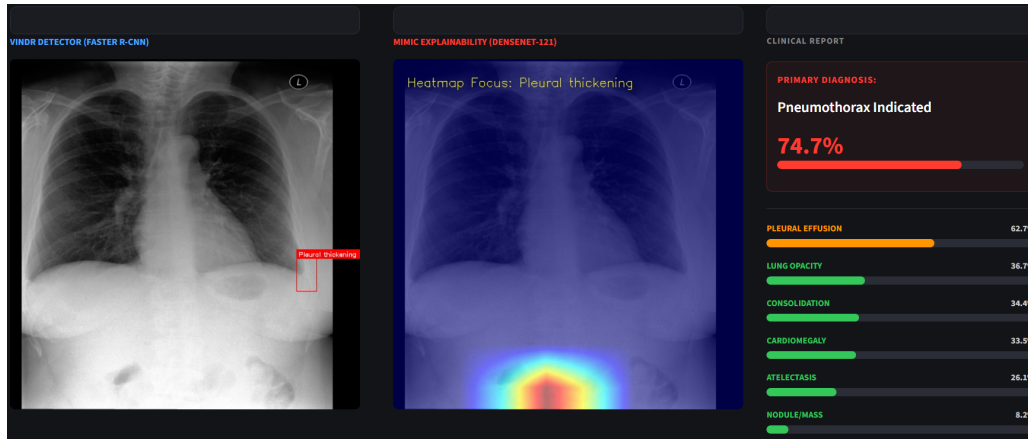


Fig. 4. The deployed MED-AI Visualizer on a deliberate failure case. The detector localizes pleural thickening on the left lateral margin while the classifier names pneumothorax at 74.7%, and the saliency sits sub-diaphragmatic rather than over the pleural margin.

The mean AUC-ROC of 0.759 is below the published range, and two contributing factors were identified: per-class ranking degrades to near-chance on small focal findings, and top-1 predictions concentrate on five of fourteen categories.

Four directions follow. The first is full-corpus retraining at higher resolution with probability calibration, which should account for most of the AUC gap. The second is weighted box fusion [12] in place of greedy NMS, with transformer-based detectors evaluated alongside. The third is joint multi-task optimization over a harmonized ontology, so that localization supervision can regularize the classifier. The fourth is a prospective reader study to determine whether saliency overlays alter radiologist decisions, which would be needed to establish clinical value.

ETHICS AND DATA AVAILABILITY

Every image used here comes from an openly distributed, de-identified, retrospectively assembled collection; no participant was recruited prospectively for this work. Access to MIMIC-CXR [15] runs through PhysioNet under its credentialed-access terms, and its images carry HIPAA Safe Harbor de-identification. VinDr-CXR [7] was gathered with Institutional Review Board approval. At no point was any individual re-identified. Trained weights and source code can be obtained from the authors on request. We are grateful to the curators of both collections, and to the radiologists whose reading underpins the annotations.

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